

If an inode has size = 600 bytes and block size = 512 bytes, how many bytes are stored in the second block?

- A. 512
 - B. 600
 - C. 100
 - D. **88**
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If a filesystem runs out of inodes but has free disk space, what error occurs?

- A. Segmentation fault
 - B. Kernel panic
 - C. **"No space left on device"**
 - D. Permission denied
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In singly indirect addressing, if block size = 4096 bytes and pointer size = 4 bytes, how many data blocks can be referenced?

- A. 256
 - B. 512
 - C. **1024**
 - D. 2048
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During kernel boot in Lab 0, the kernel panic occurs because:

- A. CPU not initialized
 - B. RAM missing
 - C. **No root filesystem (no init process available)**
 - D. Disk corrupted
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Why is block size an important design decision?

- A. Affects CPU speed
 - B. Tradeoff between I/O efficiency and fragmentation**
 - C. Affects RAM only
 - D. No impact
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Why is QEMU used in kernel labs?

- A. Faster CPU
 - B. Safe environment to test kernel without crashing host system**
 - C. Better graphics
 - D. Less memory usage
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What does the system call `read(fd, buf, n)` do?

- A. Opens file
 - B. Writes data
 - C. Reads n bytes from file descriptor into buffer**
 - D. Closes file
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Internal fragmentation refers to:

- A. Wasted space inside blocks**
 - B. Free space exists but not in a contiguous region
 - C. File corruption
 - D. Memory leaks
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In contiguous allocation, why is sequential file access fast?

- A. Metadata is smaller
 - B. Data blocks are stored next to each other on disk**
 - C. Files are compressed
 - D. CPU caches data
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Why must BusyBox be compiled as a static binary in early boot stages?

- A. For faster execution
 - B. Because shared libraries are not yet available in the filesystem**
 - C. To reduce memory usage
 - D. To avoid kernel calls
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